



mobium

The app said it was playing. **Nothing was heard.**

How to test what a mobile app plays: hear it, assert it, and see what cut across it. Measured on an Android emulator and a Pixel 8 Pro.

```
playing: silence for 2 s
```

```
the app
```

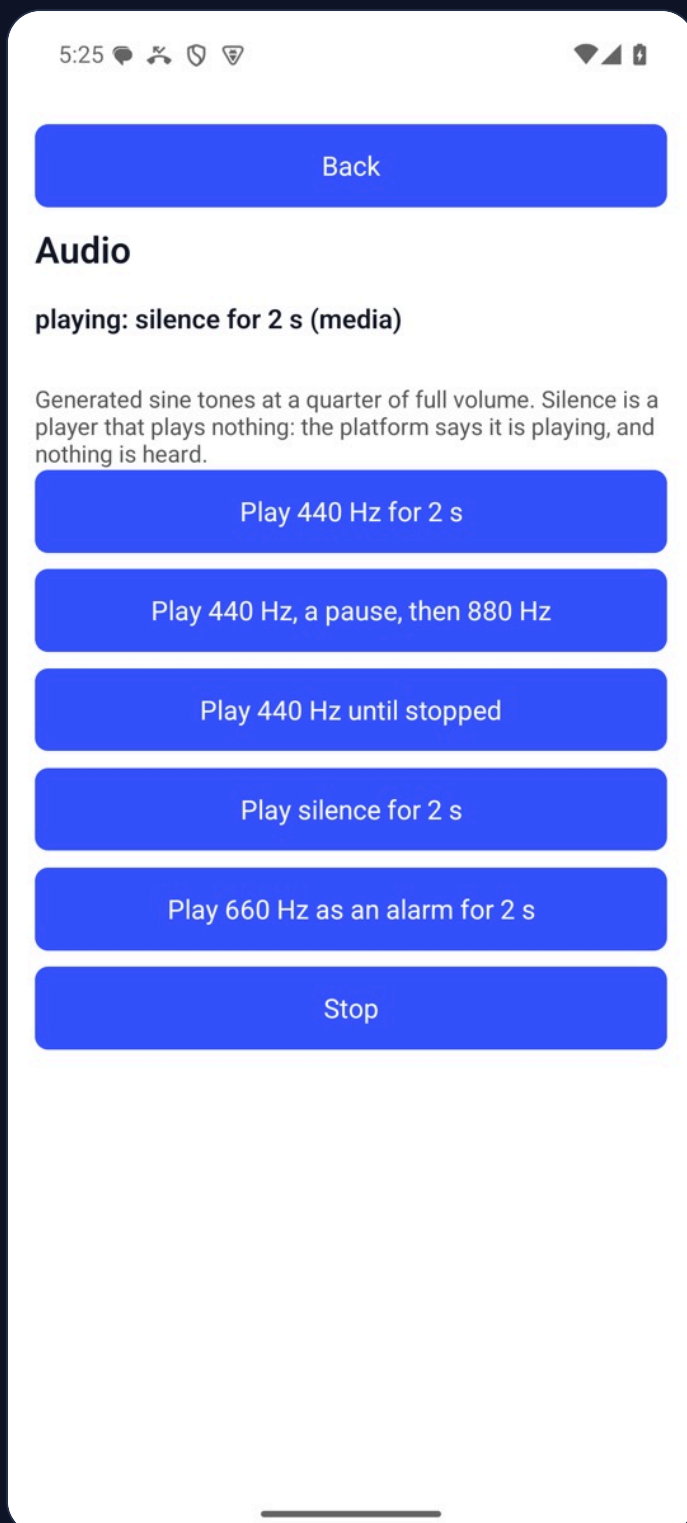
```
state: started
```

```
Android
```

```
silence throughout
```

```
the capture
```

Everyone said it was playing.



MobiumApp's Audio Demo plays two seconds of zeros: a player that is started, and silent.

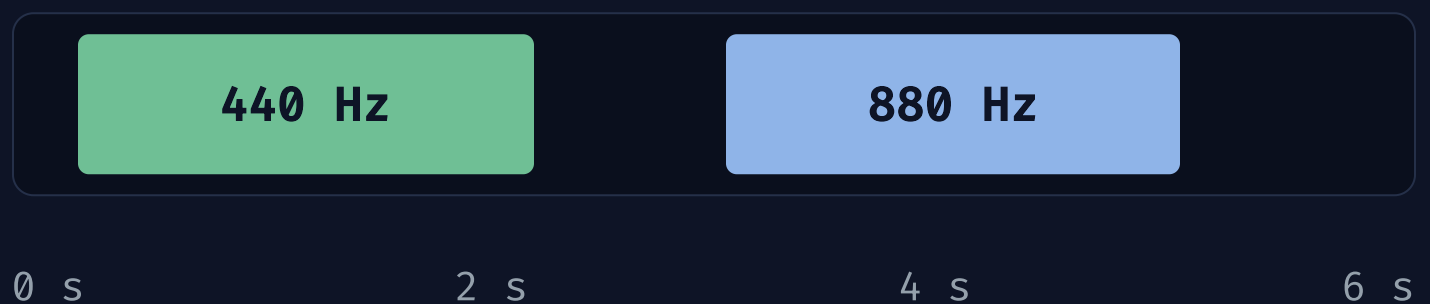
The app's own line says **playing**. Android's audio service says the player is **started** — exactly what it says for a tone.

Neither can tell you whether anything was heard. Only the sound can.

A timeline, not a waveform.

```
$ mobium audio start  
$ mobium tap testid=audioSequence  
$ mobium audio stop -o sequence.wav
```

heard



Sound and silence to a tenth of a second, each sound's pitch and level, read from the emulator's own audio. Nothing is installed on the device.

440 Hz, a second of silence, 880 Hz

Pitch and length, in order.

```
$ mobium audio stop -o seq.wav --expect 440:1.8-  
2.3,880:1.8-2.3  
saved seq.wav: ... 440 Hz ... 880 Hz ...
```

```
$ mobium audio stop -o tone.wav --expect 880  
error: expected 880 Hz; heard 440 Hz for 2.1s  
(0.3-2.4s) – the capture is saved at tone.wav
```

A wrong expectation fails, says what was heard, and keeps the capture as the evidence. **--expect silence** asserts nothing was heard; a tap's click, a tenth of a second, does not count.

One step in a test file.

```
{"audio": {"action": "stop", "expect": [  
  {"hz": 440, "min_ms": 1800, "max_ms": 2300},  
  {"hz": 880, "min_ms": 1800, "max_ms": 2300}  
]}}
```

```
$ mobium test tests/heard.test.json  
2 passed  
$ mobium test tests/wrong.test.json  
0 passed, 2 failed # each saying what was heard
```

The same from Python, Java, JavaScript, Go and .NET.

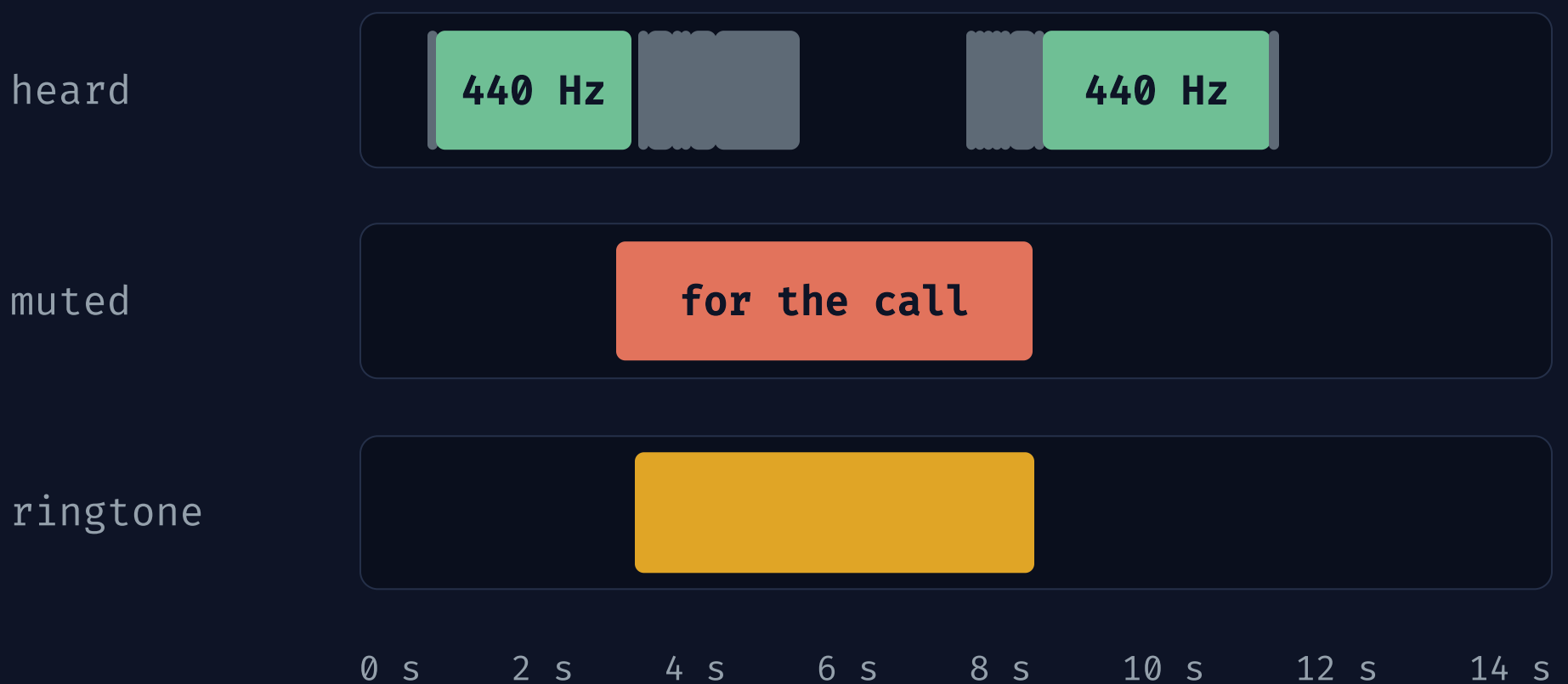
What arrives follows the volume.

The same tone, at four media volumes:

MEDIA VOLUME	HEARD AT
15 of 15	-9 dBFS
5 of 15	-42 dBFS
1 of 15	-62 dBFS
0 of 15	silence

So a test asserts pitch and length, never level – and every answer says the volume it was taken at: **“at media volume 0 of 15, where nothing played as media is heard”**.

A call came in. The app never knew.



Android silenced the app for as long as the phone rang and gave it back after. The app said it was playing throughout. The answer says what happened:

```
interrupted: muted for a call 3.0–8.0 s,  
a ringtone played 3.3–8.0 s
```

Any sound that cuts across the app.

The app muted

for a call, or by the device's or its own volume — with Android's reason.

Another app's sound over it

a ringtone, an alarm, a notification, a voice call, the assistant, navigation, another app's media — each named by what it is for.

A Clock timer's alarm over the tone

emulator: **an alarm played 4.4–6.5 s**

Pixel 8 Pro: **an alarm played 5.3–10.5 s**

Read from Android's audio service. A touch's click is not an interruption.

A phone's sound stays on the phone.

Nothing outside a phone hears what it plays. On an Android phone a capture records what interrupted the app instead — the same answer, with nothing installed:

```
$ mobium audio stop  
its sound is not captured on a phone;  
interrupted: an alarm played 5.3–10.5 s
```

Any phone adb reaches

a device farm's included. Making one ring takes a call from outside, which only a farm with a line can place.

An emulator hears everything

and rings on demand, with nothing else to set up.



mobium

Hear your app in a few minutes.

```
$ go install  
github.com/mobiumdev/mobium/cmd/mobium@main  
$ mobium audio start  
$ mobium audio stop -o heard.wav --expect 440
```

MOBIUM

github.com/mobiumdev/mobium

QUICK START, IN TEN MINUTES

github.com/lana-20/mobium-audio-demo

THE TALK, WITH THE CAPTURED SOUND

lana-20.github.io/mobium-audio-demo